

Multiple Choice

1. **Answer:** B

Options: A) hydrogen bonds; B) free radicals; C) allotropes; D) isotopes; E) nonpolar molecules

Note: Correct option: B - free radicals

2. **Answer:** A

Options: A) O₂; B) NO₃; C) NO₂; D) CO₃²⁻; E) NO₂

Note: Correct option: A - O₂

3. **Answer:** A

Options: A) 310.2 MHz; B) 500.5 MHz; C) 275.0 MHz; D) 400.0 MHz; E) 120.2 MHz

Note: Correct option: A - 310.2 MHz

4. **Answer:** A

Options: A) very electronegative; B) typically metallic; C) highly colored; D) mostly gases at room temperature; E) often very reactive with water

Note: Correct option: A - very electronegative

5. **Answer:** A

Options: A) H₂Se; B) H₂CO₃; C) H₂S; D) HBr; E) H₂O

Note: Correct option: A - H₂Se

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(a) 2.1 eV and (b) 48.4 Kcal/mole'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The cold water causes a decrease in the swimmer's metabolic rate.'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $\text{Na}_2\text{SO}_3 + \text{FeCl}_2$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 4.5 R. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.030 V. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key